Bachelor's Thesis

# KAIZEN STRATEGIES FOR IMPROVING PLANNING AND PRODUCTION FLOW IN PHARMACEUTICAL ENVIRONMENTS

**Double degree in Chemical Engineering and Industrial Electronics and Automation.**

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**Report and Annex**

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## Glossary

**Eighty-twenty Rule** Refers to the Pareto principle, which states that roughly 80% of effects come from 20% of the causes. 13

**Gemba** A Japanese word meaning "the real place." In a KAIZEN™ context, it usually refers to the place where value is added, such as the shop floor. It can also refer to any place where work is being performed. 7

**GQCDM Principles** An acronym for Growth, Quality, Cost, Delivery, and Motivation. It represents a strategic framework used in business and manufacturing to guide performance improvements.. 10, 17

**JIT** Just In Time. A system that "pulls" parts through production based on customer demand rather than "pushing" them based on projected demand. It results in line balancing and little to no excess inventory. 6, 7

**KAIZEN™** A Japanese term meaning "change for the better." It is a gradual, long-term approach that focuses on small, incremental changes to improve efficiency and quality. It was popularized by Masaaki Imai. 6–8, 10, 11, 16, 17

**KPI** An English acronym for Key Performance Indicators, which are the key indicators of a given process or activity. 8, 9, 11

**Lean** An English term meaning "no fat." In a management context, it is a strategy to reduce or eliminate waste or excess. 6, 7

**Muda** A Japanese word for "Waste" and a key concept in TPS. It is one of three types of deviation from optimal resource allocation (Muda, Mura, and Muri). 6, 7, 11

**OPL** Stands for One Page Lesson. Is a simple, visual, and concise training document used on the shop floor to quickly and effectively transfer knowledge. The key characteristic of an OPL is its brevity; it contains all the necessary information for a specific task or procedure on a single page. OPLs are typically created and maintained by the operators themselves, ensuring the content is practical and relevant. They are a core tool in Lean Manufacturing for standardizing best practices, providing quick reference for machine setup or troubleshooting, and promoting continuous improvement. They serve as a powerful alternative to lengthy manuals, making training accessible and immediate.. 14

**SMED** Stands for **Single-Minute Exchange of Die**. A Lean manufacturing methodology for drastically reducing the time it takes to complete equipment changeovers. The goal is to make changeovers so quick that it becomes economically viable to produce smaller batch sizes, reducing inventory and improving production flexibility.. 13, 14, 16

**TPS** Toyota Production System. A methodology that resulted from over 50 years of KAIZEN™ at Toyota. It is built on a foundation of Leveling with supporting pillars of Just-in-Time and Jidoka. 6

**Visual Management** A set of techniques for creating a workplace that uses visual communication and control. The philosophy is "what gets measured and displayed, gets done." Simple tools are used to identify the target state and correct any deviations. 7

**VSA** Stands for Value Stream Analysis. A Lean management method used to visually map and analyze the flow of materials and information required to deliver a product or service to a customer. VSA identifies value-added and non-value-added activities, highlights waste, and provides a basis for Kaizen (continuous improvement) efforts. It typically includes a current-state map, future-state vision, and an actionable plan to close gaps.. 12, 16

# Contents



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# 5 Preface

## 5.1 The KAIZEN™ Philosophy

After World War II, Toyoda Kiichiro, then president of the Toyota Motor Company, saw the need to catch up with America to assure the survival of the automobile industry of Japan. Taiichi Ohno, Toyota's engineer, had a challenge: how to build a diversified line of products with the limited equipment Toyota possessed at the time. Instead of using batch production, he designed a system of integrated production that used one-piece flow, demonstrating that a trade-off between quality and productivity does not need to exist (Netland and Powell, 2016). Aligned with other practices, Toyota Production System (TPS) was born to defy the premise that bigger batches mean lower costs.

TPS, which ultimate goal is the absolute elimination of waste, is based in two pillars: just-in-time (JIT) and autonomation (automation with a human touch). The lack of the necessary capital resources to support high inventory levels was solved with the implementation of a just-in-time pull system. JIT means "to produce the necessary units at the necessary quantities at the necessary time". A company establishing this flow throughout can approach zero inventory. With autonomation, Toyota intended to create a built-in automatic checking system against small abnormalities. With this prevention mechanism, defective products were not produced. That was not the only advantage. While the machine is working properly, employees are liberated. Consequently, one worker could attend multiple machines, increasing production efficiency. Furthermore, autonoma- tion came as a facilitator of improvement. Stopping the machine creates awareness and, through the total comprehension of the problem, improvement measures can be taken. (Ohno, 1988) Lean, based on TPS, is a methodology that accentuates customer needs, to improve quality and reduce costs through continuous improvement and employee involvement (Graban, 2008). KAIZEN™, a Japanese word, refers to the concept of continuous improvement. It does not intend to be mistaken as innovation or disruption and elevated costs. On the other hand, KAIZEN™ is about small and subtle improvements. It is a low-cost low-risk process that assures incremental progress and sustainable changes in the long term.

At the beginning of the 21st century, Toyota Motor Company surpassed General Motors and became the world leader in automobile production, expanding and elevating the KAIZEN™ concept for its key role in the company's success. KAIZEN™'s methodology is based on five fundamental principles described in Massaki and Sanchez's book [1] :

- Create customer value

  Value is what the customer is willing to pay for. Using a market-in approach to make informed decisions based on what the customers want and offer them that in the best way possible, improving customer experience.

- Create flow efficiency

  Flow efficiency can be obtained through the elimination of three Ms: Muda (waste), Mura (unevenness) and Muri(overburden). The concept of Muda primarily originated from Taiichi Ohno's production philosophy in the early 1950s. He defined, in the industrial context he was inserted, seven sources of Muda: overproduction, waiting, inventory, motion, transportation, over-processing and defects. All the activities that do not add value to the process are considered Muda. Fujio Cho, former President of Toyota, defined waste as "anything other than the minimum amount of equipment, materials, parts, space and worker's time, which are absolutely essential to add value to the product". Eliminating waste is the most effective way to increase productivity and decrease costs. Although Muda has gained bigger awareness, comprehending all three concepts allows for a greater understanding of Lean (Netland and Powell, 2016). Mura is the variation in a process that is not caused by the final customer, for example highs and lows in the planned production due to the production system or the irregular work rhythm. Leveling production using JIT can eliminate this irregularities, avoiding work spikes and long waiting times for workers. Muri means to overburden equipment or workers, demanding a faster rhythm for a longer period than expected. While in equipment it can provoke defects and failures, in people it can result in safety issues.

- Be Gemba oriented

  Gemba is the place where value is added. Problems should be identified and solved there, at their root cause. Central to KAIZEN™ is *genchi genbutsu* (go and see for yourself). Going to the Gemba means going to the place where the action really happens, to be able to thoroughly observe the reality of the processes.

- Empower people

  Respect for people is a fundamental principle of the KAIZEN™ philosophy. Kaizen recognizes that people are not the problem, processes are. It intends to improve every day, everywhere, with everyone, training all employees and encouraging people to think and share their own improvement ideas, systematically. It is in management's biggest interest to create the best possible conditions for everyone to do the best possible work.

- Be scientific and transparent

  As important as identifying problems, is knowing how to explain and substantiate one's findings, supporting them with data. Visual Management is a basal tool of KAIZEN™, that highlights problems and allows for a quicker reaction to them.

# 6 The Problem at hand

## 6.1 Introducing the company

Now that we have introduce the base methodology of KAIZEN™ we must understand the focus of this work. The project describe has been implemented in a Pharmacology Company located on the Catalonia area. It focuses primarily on providing Contract Manufacturing and Packaging Services for Medicines and Dietary Supplements, both for Human and Veterinary use.

The facilities are equipped with state-of-the-art machinery and have a total area of 7.200 $m^2$, distributed across two facilities, one dedicated to the manufacturing of pharmaceuticals and the other to nutraceuticals.

In total the company has over 400 employees and about 80% of the workforce is established on the shop floor.

There are several manufacturing areas and work lines:

- TMP (Raw materials handling):

  This section is where the mixture of active components, either liquids or solids, and ingredients are composed.

- PI (Compression):

  In this section compression of solids into tablets occurs.

- PAC (Conditioning):

  In this section the formula is conditioned. There are 3 forms of presentation:

  Bottles: Only for liquids.

  Jars: Containers of tablets for solids.

  Pouches: For single use uncompress solids.

## 6.2 Scope and pilots

The scope of this project can be narrow down into 3 *KAIZEN™ Events*, each one will be implemented in a different work line as a pilot.

Each KAIZEN™ Event will be:

- Improvement of Line Efficiency.

  Pilot: "Jars" Line

  KPI: Line OEE

- Improvement of line Productivity

  Pilot: "Liquids" Line

  KPI: Line Productivity

- Improvement of planning

  Pilot: Sourcing and production planning of PAC area.

  KPI: OTIF

- Creating a system for daily management and continuous improvement culture.

  Pilot: "Jars" Line

  KPI: Economic impact of meetings sourced improvement ideas.

# 7 Improvement of Line Efficiency

## 7.1 Objective Function

As in any optimization problem, we must first define the objective function. In the KAIZEN™ methodology, improvement efforts are directed toward what is most valuable for the client, often summarized by the principles of GQCDM Principles:

- **Growth**: Focus on expanding market share, innovation, and business development.
- **Quality**: Ensuring high standards in products, processes, and customer satisfaction.
- **Cost**: Controlling expenses and improving efficiency to remain competitive.
- **Delivery**: Meeting deadlines and ensuring timely, reliable delivery of products or services.
- **Motivation**: Encouraging and empowering employees to foster engagement, productivity, and innovation.

In this particular KAIZEN™ event, the focus will be on minimizing **Production Cost**. Specifically, this is pursued by maximizing the sustainable production capacity of the line.

$$\text{Objective function} = \max\left(\frac{\text{Total output units}}{\text{Time window}}\right) \tag{1}$$

In other words, equation 1 seeks to maximize the throughput rate, for example the number of orders the factory can process in a year.

This can also be expressed as:

$$\text{Objective function} = \max\left(\text{Rated line speed} \cdot OEE\right) \tag{2}$$

Here, the rated line speed represents the theoretical design capacity, while $OEE$ adjusts for availability, performance, and quality losses. The introduction of this efficiency factor will be discussed in the following section.

It is evident that production output is strongly embedded in the cost structure of a company. A significant share of manufacturing cost is related to the utilization of machines and operators. By producing more in less time, the cost per unit decreases, thereby improving overall competitiveness.

## 7.2 OEE and profits calculation method

Overall Equipment Effectiveness (OEE) is a standard metric used to measure the efficiency of a production line. It identifies how much of the planned production time is truly productive. OEE is calculated by multiplying three core components: Availability, Performance, and Quality.

### 7.2.1 Availability

Availability measures the proportion of scheduled production time during which equipment is operational and ready for use. It accounts for planned downtime, such as maintenance or shift breaks, as well as unplanned losses caused by equipment breakdowns or delays.[2]

$$A = \frac{\text{Operating Time (h)}}{\text{Planned Time (h)}} \tag{3}$$

With that definition we first have to acknowledge 2 main terms:

1. Planned Time: The total scheduled production time.
2. Operating time: The Planned Time minus all planned stoppages like changeovers, cleaning, and scheduled breaks.

### 7.2.2 Performance

Performance evaluates whether equipment is running at its optimal speed during production. It captures losses due to reduced speed or minor stoppages that may not halt production but reduce efficiency.[2]

$$\mu = \frac{\text{Actual Units Produced (units)}}{\text{Operating Time (h)} \cdot \text{Nominal Speed (units/h)}} \tag{4}$$

### 7.2.3 Quality

Quality assesses the proportion of defect-free products produced relative to the total output. This metric highlights losses due to rework or defective items, which reduce the overall effectiveness of the production process.[2]

$$Q = \frac{\text{Total OK unit output (units)}}{\text{Actual Units Produced (units)}} \tag{5}$$

### 7.2.4 OEE calculation

By combining these three components, we get the final OEE formula, which shows the overall efficiency of the production line.

$$OEE = A \cdot \mu \cdot Q \tag{6}$$

An OEE score of 100% represents perfect equipment effectiveness, with no downtime, optimal performance, and zero defects.[3]

Or in fair terms:

$$\begin{aligned} OEE = \frac{\cancel{\text{Operating Time (h)}}}{\text{Planned Time (h)}} \cdot \frac{\cancel{\text{Actual Units Produced (units)}}}{\cancel{\text{Operating Time (h)}} \cdot \text{Nominal Speed (units/h)}} \cdot \\ \cdot \frac{\text{Total OK unit output (units)}}{\cancel{\text{Actual Units Produced (units)}}} = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \cdot \text{Nominal Speed (units/h)}} \end{aligned} \tag{7}$$

$$OEE = A \cdot \mu \cdot Q = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \cdot \text{Nominal Speed (units/h)}} \tag{8}$$

### 7.2.5 Profit Calculation

Now that the concept of $OEE$ is clear we could go back to equation 2. In KAIZEN™ philosophy we focus on the non investment improvements. As we can infer changing the Nominal Speed of a line or machine could mean investing a huge amount of money in technology, instead of focusing on eliminating Muda. That's why we will focus on OEE as a KPI as the main metric for this project.

We can quantify the augmented capacity derived from an improvement in OEE as the reduced planned operating time of the line. In this work we will not be focused on improving sales directly, so the Total Number of units produced will be fixed. We wont focus on investing in new equipment, so the Nominal Speed of the line will stay constant. Every point of improvement of OEE will be inversely related to a decreased in the need for more planned time which in turns decreases the operating cost of labor of the line. Relating back to the objective function, equation 1.

## 7.3 Identifying improvement opportunities

A comprehensive analysis of Overall Equipment Effectiveness (OEE) provides a structured approach to identifying production inefficiencies. Following a VSA, our project focused on the key opportunities identified to improve OEE. The project's baseline OEE was established after one month of observation in February, with the following results:

- Availability=41.6%
- Performance=66.2%
- Quality=99.89%
- OEE=27.5%

The baseline data clearly indicates that low availability is the primary factor limiting overall equipment effectiveness. This is further explained in Figure 1, which shows that only 40% of the available time is utilized for production.

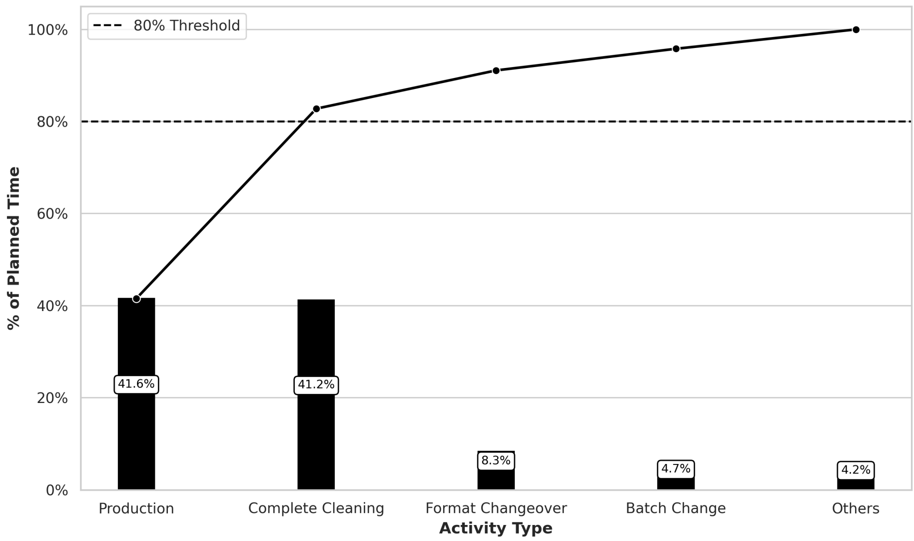


Figure 1: Pareto Chart of Planned Time dedication by Activity

The main reason for this loss in availability is the time dedicated to cleaning all machine equipment and the production space. This extensive cleaning is a direct consequence of stringent quality controls within the pharmaceutical industry, which mandate:

- The prevention of cross-contamination between different products or batches.

- The removal of product from machinery overnight to prevent quality degradation.

Figure 2, provides a more detailed breakdown of the availability losses. By applying Eighty-twenty Rule we can focus on the key areas for improvement.

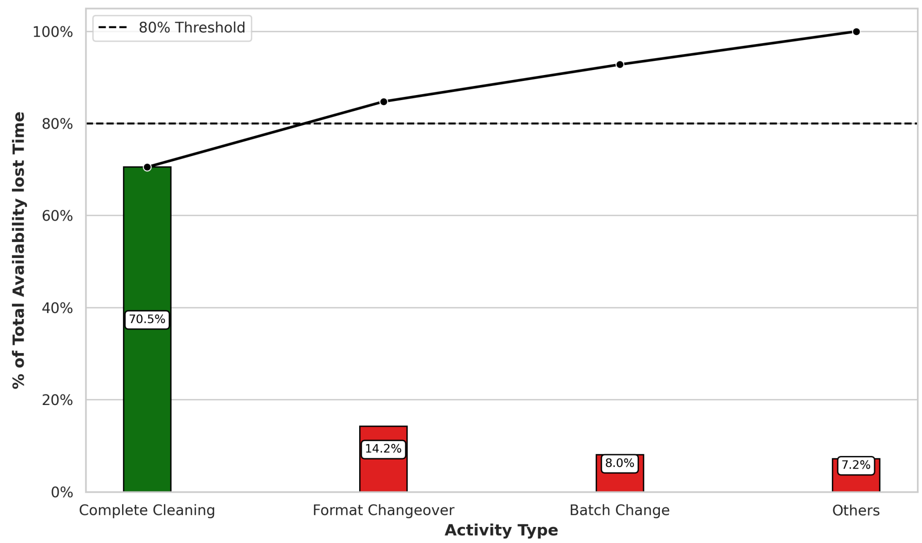


Figure 2: Pareto Chart of Main Losses on Availability. In Green Above 80% of cumulated time, in Red the rest 20% of time

The analysis confirms that a significant efficiency improvement can be achieved by reducing the time spent on cleaning. This objective will be addressed using the SMED Methodology.

## 7.4 Implementing the SMED Methodology

### 7.4.1 Brief introduction to the SMED

The SMED is a technique that helps us distinguish between Internal and External tasks as described in Dillon and Shingo [4] book *A Revolution in Manufacturing: The SMED System*. The core principle of the SMED system is to significantly reduce the time required for machine changeovers. This is achieved by converting as many setup operations as possible from *internal* tasks (which can only be performed when the machine is stopped) to *external* tasks (which can be completed while the machine is still running). The goal is to make all changeovers last less than ten minutes, thereby maximizing production time and enabling smaller, more flexible batch sizes.

This core principle of *internal* vs *external* tasks can be applied to multiple operations, not only limited to changeovers. In this case we will apply it to Cleanings. The basics steps described in Dillon and Shingo book are as follows:

1. Identify all tasks performed during the operation.
2. Identify all participants of the operation.
3. Distinguish between *internal* and *external* tasks.
4. Convert as much tasks from *internal* to *external*.

5. Distribute and balance internal tasks in order to minimize machine stopping time.
6. Standardize procedure, create checklists and OPL.
7. Measure progress and identify bottle necks.

### 7.4.2 Brief introduction to the SMED

The SMED is a technique that helps us to significantly reduce the time required for machine changeovers and setups. Developed by Shigeo Shingo in the 1950s, the core principle is to convert as many setup operations as possible from *internal* tasks (those that can only be performed when the machine is stopped) to *external* tasks (those that can be completed while the machine is still running). The ultimate goal is to complete changeovers in less than ten minutes (a "single minute" in the literal sense of the word), thereby maximizing production time and enabling smaller, more flexible batch sizes.

As detailed in Shingo's foundational book, *A Revolution in Manufacturing: The SMED System* [4], this methodology is not limited to product changeovers but can be applied to any process with internal and external tasks, such as the cleaning procedure we have identified. The SMED implementation process follows these distinct steps:

1. **Preliminary Stage**: Identify and analyze the current setup or cleaning process. This involves documenting all tasks and their timings.
2. **Stage 1: Separate Internal and External Tasks**: Critically evaluate each step of the process and categorize it as either internal or external.
3. **Stage 2: Convert Internal from External Tasks**: This is the most crucial phase. Brainstorm and implement changes that allow internal tasks to be performed while the machine is running. For example, blowing off compressed air while the line is up.
4. **Stage 3: Streamline All Remaining Tasks**: Once the conversion is complete, focus on improving the efficiency of the remaining internal and external tasks. This involves using tools like checklists, visual aids, and improved tool organization. For instance, using quick-release clamps instead of bolts.
5. **Sustain and Standardize**: Create and maintain standardized work instructions, checklists and OPLs to lock in the improvements. Establish a system for measuring and tracking progress to identify future opportunities.

This structured approach ensures a systematic and measurable reduction in changeover or cleaning time, which directly increases equipment availability.

### 7.4.3 Documentation of Tasks and Timings

The first preliminary step of the SMED methodology is to conduct a thorough analysis of the existing changeover process to establish a baseline. This was achieved through direct observation and detailed time studies, utilizing video analysis to capture every task performed by the operators and the shift leader inputs to calculate their precise timings. This meticulous documentation serves to eliminate assumptions and reveal previously unnoticed process inefficiencies. A summary of the identified tasks is presented in Table 1, while the complete, detailed list can be found in the Appendix A.

Table 1: Cleaning Tasks Sorted by Duration (Descending)

| Task | Time (min) | Task | Time (min) |
|---|---|---|---|
| Clean technical area | 120 | Clean secondary equipment | 60 |
| Clean ceilings and walls (secondary) | 45 | Clean positioner | 45 |
| Clean secondary floor | 30 | Clean Cremer 1 | 30 |
| Clean ceilings and walls | 30 | Remove leftover material | 25 |
| Clean T-Delta | 23 | Clean Cremer 2 | 23 |
| Dismantle Cremer suction equipment | 15 | Take carts to washing area | 15 |
| Soap cart 1 parts | 15 | Clean floor | 15 |
| Clear production line | 13 | Vacuum and blow | 12 |
| Apply disinfectant cart 1 | 10 | Dismantle positioner | 10 |
| Dismantle Cremer 1 | 10 | Dismantle T-Delta | 10 |
| Count materials | 10 | Bag and label clean parts | 8 |

#### 7.4.4 Separating Internal from External Tasks

With a comprehensive understanding of the current state, the team's focus shifted to categorizing each task. Through collaborative workshops and direct consultation with the operators we classified each task as either *internal* or *external*. This classification was visually represented in an *E-I* chart, which laid the foundation for the subsequent conversion efforts. The resulting categorization is presented in Table 2.

Table 2: SMED Tasks Classified as Internal and External

| Internal Task | Time (min) | External Task | Time (min) |
|---|---|---|---|
| Clean secondary equipment | 60 | Clean technical area | 120 |
| Clean positioner | 45 | Clean ceilings and walls (secondary) | 45 |
| Clean Cremer 1 | 30 | Clean secondary floor | 30 |
| Clean ceilings and walls (primary) | 30 | Remove leftover material | 25 |
| Clean T-Delta | 23 | Clean floor | 15 |
| Clean Cremer 2 | 23 | Vacuum and blow | 12 |
| Dismantle Cremer suction equipment | 15 | Count materials | 10 |
| Take carts to washing area | 15 | Bag and label clean parts | 8 |
| Soap cart 1 parts | 15 | | |
| Clear production line | 13 | | |
| Apply disinfectant cart 1 | 10 | | |
| Dismantle positioner | 10 | | |
| Dismantle Cremer 1 | 10 | | |
| Dismantle T-Delta | 10 | | |

This analysis revealed a significant opportunity for improvement. The total calculated internal time needed was a maximum of **3h 18min**, within a total cleaning time baseline of **5h 5min**. This gap represented the potential for substantial time savings. A visual representation of all operations and their timings is presented in the Appendix A, Figure 7.

#### 7.4.5 Standardizing the New Procedure

To lock in the improvements and ensure their sustainability, a new standard operating procedure was developed. This was not merely a document but a comprehensive set of visual tools created in close collaboration with the operators. Every step was detailed in a clear checklist to provide a quick visual reference for the entire process.

The new procedure was treated as a living document. Any deviation from the objective time was immediately treated as a problem to be solved. This triggered a deviation where the shift leader and team would analyze the root cause of the delay, identify a countermeasure, and update the standard procedure to prevent the issue from reoccurring. This feedback loop ensured continuous improvement and maintained the gains achieved through the SMED effort.

Also the new checklist of operation added the differentiation between internal and external task, to make sure the machine availability would be hindered as less as possible form the cleaning procedure.

An example of the checklist used can be found in the Appendix A on Figure 8.

#### 7.4.6 Results and Process Evolution

The baseline data, collected during my observations in February 2025, established the average time for the Jars line cleaning procedure at **5 hours and 5 minutes**. This total time was composed of **3 hours and 18 minutes** of internal operations (performed with the line stopped) and **1 hour and 47 minutes** of external operations that were also conducted during downtime.

Based on this analysis, a two-pronged strategic goal was established. The first and most immediate objective was to convert the **1 hour and 47 minutes** of external operations, which were previously performed with the machine off, to be conducted in parallel with production or during the preparation phase. This would immediately reduce machine downtime from 5 hours and 5 minutes to a maximum of 3 hours and 18 minutes, matching the internal operation time. The second objective, informed by the VSA analysis, was to systematically reduce the remaining internal time by a third, setting an ambitious final target of **2 hours and 15 minutes** for the entire cleaning procedure.

The evolution of the cleaning process time is visually documented in Figure 3, which tracks the procedure's duration over several weeks.

The data reveals a clear and immediate reduction in downtime following the introduction of the new Standard Operating Procedure and the strategic conversion of external tasks. Machine inactivity levels dropped drastically, reflecting the direct impact of the SMED methodology. The figure also demonstrates a clear downward trend in cleaning time over the subsequent months, indicating a culture of continuous improvement and effective knowledge sharing among the operators. This sustained improvement is a powerful testament to the successful application of KAIZEN™ principles beyond the initial project scope.

A good example of this came in week 24, when we introduced a new, new, high-viscosity product tha disrupted established processes. At first, the adhesive characteristics proposed an unexpected challenge threatening the steady progress on the adaptation of the new standard. However, the team quickly used the standard procedure and checklist as a guide to break the problem down and adjust on the fly. Operators worked together to test ideas, tweak tools and cleaning products. Settling on a better method. By the next week, they had updated the standard and were back on track with our time targets. This showed just how adaptable the team had become and how a strong Kaizen culture helps us solve problems quickly and lock in improvements.

As a conclusion, from the first observation of a Cleaning machine downtime of **5h 5min** to the final observation right before summer break of **1h 36min** there is an improvement of **68%** reduction on machine downtime during Cleaning procedures.

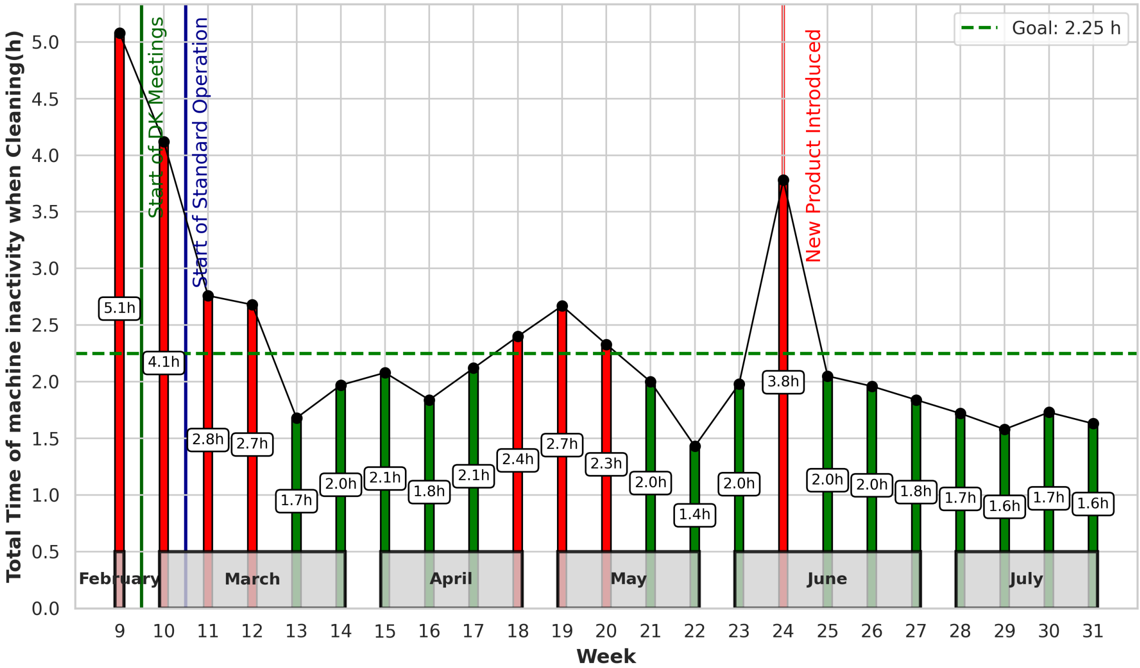


Figure 3: Evolution of the Cleaning Time Procedure along the Weeks.

# 8 Improving Line Productivity

## 8.1 Objective Function

This KAIZEN™ event is grounded in GQCDM Principles principles, with a primary focus on enhancing line productivity through the optimal allocation of human resources necessary to complete a production batch. The analytical perspective shifts from traditional machine efficiency metrics to the evaluation of human time investment, identifying labor allocation as a critical lever for productivity improvement.

The objective function for this Kaizen event is formally defined as the maximization of productivity, as expressed by the following equation:

$$\text{Objective Function} = \max\left(\text{Productivity}\right) = \max\left(\frac{\text{Unit output (units)}}{T_{\text{operation}}, \text{(h)} \cdot N_{\text{operators}}}\right) \tag{9}$$

Where:

- *Unit output*: Represents the total number of units produced within a defined production batch.
- $T_{operation}$: Denotes the total person-hours invested in production, expressed as the sum of all operator working hours.
- $N_{operators}$: Refers to the number of operators assigned to the production line.

Maximizing productivity, as defined by this function, inherently targets production cost reduction. By producing the greatest unit output for the minimum operator time, the event directly aligns with Kaizen principles of waste elimination and resource optimization, as each operator-hour represents a significant and reducible cost.

## 8.2 Productivity and profits calculation method

### 8.2.1 Productivity

For this study, productivity will be calculated using Equation 9. Because products produced on the same line may exhibit different standard line speeds, this factor is incorporated to enable fair comparisons across different product types. For the remainder of the analysis, we refer to productivity improvements as ΔProductivity. Measured relative to the previous standard as described in Townsend's book *The basics of line balancing and JIT kitting* [5].

$$\Delta\text{Productivity}(\%) = \frac{\left(\dfrac{\text{Unit Output}_1}{T_{\text{operation},1} \cdot N_{\text{operators},1}}\right)}{\left(\dfrac{\text{Unit Output}_0}{T_{\text{operation},0} \cdot N_{\text{operators},0}}\right)} - 1 \tag{10}$$

Here, the subscript 0 refers to the baseline or historical production values, while subscript 1 corresponds to the post line-balancing measurements.

### 8.2.2 Profit Calculation

In alignment with *OEE* profit analysis, each incremental gain in productivity is translated directly into cost savings by reducing the total operation time required to produce the same number of units. This approach establishes a direct quantitative link between productivity improvements and financial performance, consistent with Lean Kaizen principles of continuous improvement and cost efficiency.

### 8.2.3 A Brief Description on Line Balancing and Yamazumi

The foundational principle of line balancing was applied to the production process with the primary objective of distributing tasks equally among operators. This approach aimed to eliminate bottlenecks and minimize idle time, thereby optimizing workflow. The key analytical tool used for this work was the Yamazumi chart. This visual method, which literally translates to "to pile up," provided a graphical representation of each operator's work content and the time required for their individual tasks. By visualizing the cumulative workload for all team members, the Yamazumi chart immediately highlighted the imbalances and bottlenecks that were impeding the process. This visual data was critical for identifying and eliminating the forms of waste known as muda, mura, and muri—specifically, the waste of waiting, unevenness, and overburden—that were inherent in the unbalanced line. Through the strategic use of the Yamazumi chart, tasks were redistributed to create a smoother, more efficient, and balanced workflow, which directly contributed to the overall objective of maximizing productivity.

In conjunction with line balancing, the *Takt Time* is a critical metric for determining the required pace of production to meet customer demand. It serves as a heartbeat for the production process, establishing a baseline against which the balanced line's performance can be measured. The formula for calculating Takt Time is defined as (11):

$$\text{Takt Time} = \frac{\text{Total Available Production Time}}{\text{Customer Demand}} \tag{11}$$

The *Takt Time* will be used to determine the number of operators needed on the line in order to cover customer demand. As an example let's look onto the planned production for one shift. Let's say it is 26,400 units that must be covered in one shift of 7.33h. In this case following equation 11, the *Takt Time* must be:

$$\text{Takt Time} = \frac{7,33h \cdot \frac{3,600s}{1h}}{26,400units} = 0.99units/s \tag{12}$$

Meaning that the line should be averaging a production of roughly 1 unit per second.

### 8.2.4 Initial Line distribution

As stated in previous sections, the pilot line for this endeavour was the "Liquids" Line of the plant. In this line drugs sold and administered in liquid form are bottled, boxed, added with the dosification device necessary, packed into distribution boxes and palletized ready for shipment.

It is divided onto 2 main section, the primary section, and the secondary. In the pharmaceutical industry, the primary section is where the product is treated outside of its shipping container, in this case its where the liquid medicine comes out from the reactor into the bottler. The primary section is regulated under heavier security measures as the product is in contact with the environment at it is more likely to be contaminated. The secondary section is where the conditioning of the bottle takes place, this is the labelling, boxing, prospect insertion, dosser insertion, and further transformation occurs. As the product is already bottled up and has no interaction with the environment the regulations are much looser. The two sections are physically separated by a positive pressure wall and to access the primary one must change his attire into a more restricted one which in turn difficult the movement between the two sections.

As depicted in the line schematics, one person must reside at all times in the primary section and as movement between the two section is heavily restricted the focus of this project will be put specially on the secondary section of the line, as not only it host the most amount of resources it is where the majority of the transformation takes place.

As shown in Figure 4 the line hosted 2 operators and 1 line coordinator.

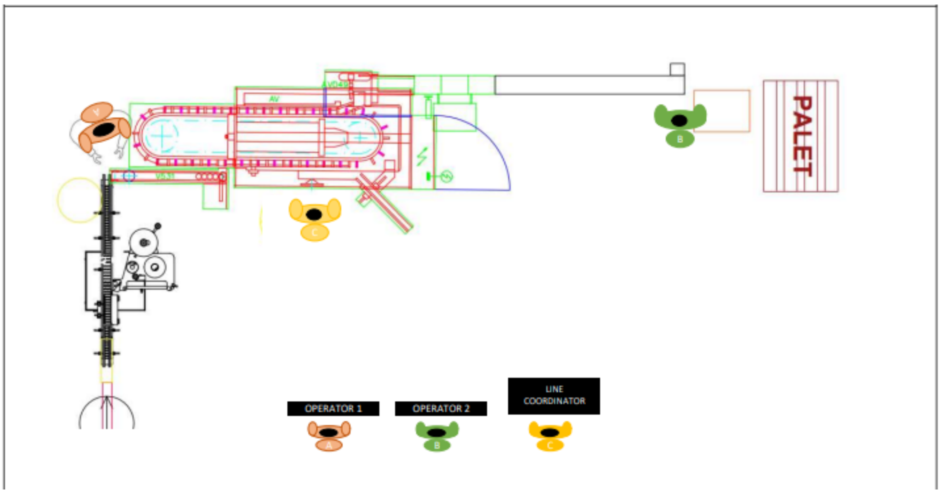


Figure 4: Line Diagram of Operator distribution

After an initial observation round. Every task done by each operator during the shift was recorded, classified and inspected. The table showing details can be found in the Appendix B Table 4.

The line *Takt Time* was calculated to 2.8s. The Yamazumi chart in figure 5 shows task distribution between operators. It is clear that operator occupation is not maximized. This chart shows us that the need for 3 operators in the line is not due to high occupation and line tasks but rather due to a lack of time organization and task sequencing.

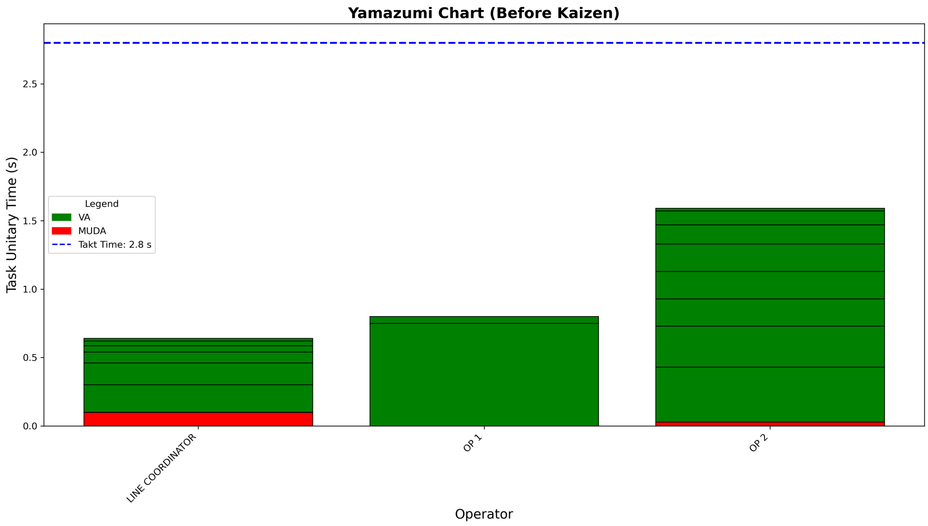


Figure 5: Yamazumi Chart Showing Task distribution before improvement.

### 8.2.5 Line Balancing

The main problem was indeed time organization. Some operations had to be performed simultaneously. Like quality checks, reloading leaflets or cases. Changing the optics from the operations, it was noticed that the bottle neck of the line did not come from the secondary section but rather how fast the primary bottling would go. So it would be not a problem to speed up the secondary cycle time and have some slack to stop the line for a few minutes in order to conduct some of this tasks, like quality controls, loading of leaflets, changing labels, change cases, etc. Meanwhile build up a buffer in the line and systematically clear it up. Maintaining this way the average line *Takt Time*.

Organizing the time in this matter and reducing the Non-added Value activities as much as possible, the line tasks where reorganized to fit only 2 operators.

The new Yamazumi chart with only 2 operators is shown in Figure **??**.

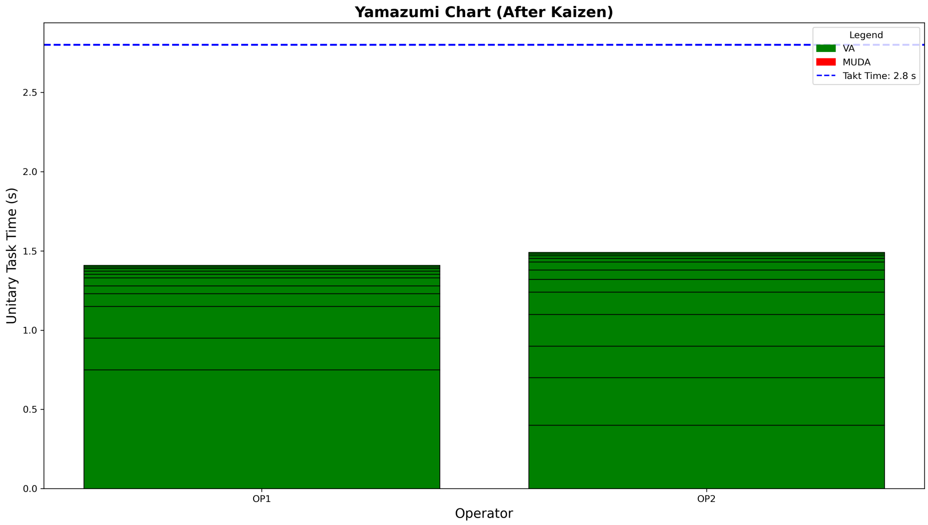


Figure 6: Yamazumi Chart Showing Task distribution with 2 operators.

# 9 References

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# Appendices

## A SMED Task Detail

Table 3: Cleaning and Production Tasks Sorted by Duration (Descending)

| **Task** | **Time (min)** | **Task** | **Time (min)** |
|---|---|---|---|
| Clean technical area | 120 | Clean secondary equipment | 60 |
| Clean ceilings and walls (secondary) | 45 | Clean positioner | 45 |
| Clean secondary floor | 30 | Clean Cremer 1 | 30 |
| Clean ceilings and walls | 30 | Remove leftover material | 25 |
| Clean T-Delta | 23 | Clean Cremer 2 | 23 |
| Dismantle Cremer suction equipment | 15 | Take carts to washing area | 15 |
| Soap cart 1 parts | 15 | Clean floor | 15 |
| Clear production line | 13 | Vacuum and blow | 12 |
| Apply disinfectant cart 1 | 10 | Dismantle positioner | 10 |
| Dismantle Cremer 1 | 10 | Dismantle T-Delta | 10 |
| Count materials | 10 | Bag and label clean parts | 8 |
| Final rinse cart 1 | 8 | Remove cleaning materials | 8 |
| Partial soaping cage 1 | 8 | Bring cleaning materials upstairs | 5 |
| Bring parts upstairs | 5 | Close logs and cards | 5 |
| Access primary area | 5 | Initial rinse cart 1 | 3 |
| Initial rinse cage 1 | 3 | Remove materials | 3 |
| Bring carts | 3 | Close guide | 3 |
| Record in production log | 3 | Change cards | 3 |
| Remove leftover material (conditioned) | 3 | Bring vacuum cleaner | 3 |
| Take cart 1 to dryer | 1 | Insert cart 1 into washer | 1 |

## B Line Balancing Details

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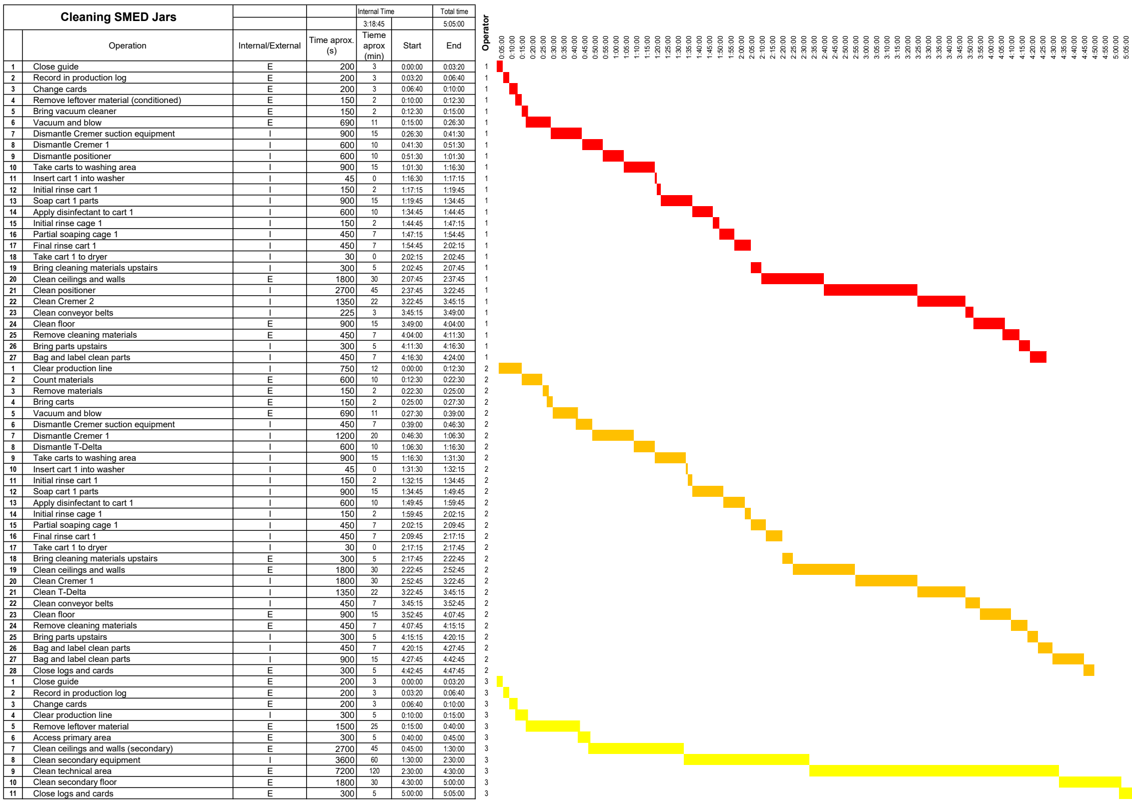

| Cleaning SMED Jars | | | | Internal Time | | Total time | |
|---|---|---|---|---|---|---|---|
| | | | | 3:18:45 | | 5:05:00 | |
| | Operation | Internal/External | Time aprox. (s) | Tieme aprox (min) | Start | End | Operator |
| 1 | Close guide | E | 200 | 3 | 0:00:00 | 0:03:20 | 1 |
| 2 | Record in production log | E | 200 | 3 | 0:03:20 | 0:06:40 | 1 |
| 3 | Change cards | E | 200 | 3 | 0:06:40 | 0:10:00 | 1 |
| 4 | Remove leftover material (conditioned) | E | 150 | 2 | 0:10:00 | 0:12:30 | 1 |
| 5 | Bring vacuum cleaner | E | 150 | 2 | 0:12:30 | 0:15:00 | 1 |
| 6 | Vacuum and blow | E | 690 | 11 | 0:15:00 | 0:26:30 | 1 |
| 7 | Dismantle Cremer suction equipment | I | 900 | 15 | 0:26:30 | 0:41:30 | 1 |
| 8 | Dismantle Cremer 1 | I | 600 | 10 | 0:41:30 | 0:51:30 | 1 |
| 9 | Dismantle positioner | I | 600 | 10 | 0:51:30 | 1:01:30 | 1 |
| 10 | Take carts to washing area | I | 900 | 15 | 1:01:30 | 1:16:30 | 1 |
| 11 | Insert cart 1 into washer | I | 45 | 0 | 1:16:30 | 1:17:15 | 1 |
| 12 | Initial rinse cart 1 | I | 150 | 2 | 1:17:15 | 1:19:45 | 1 |
| 13 | Soap cart 1 parts | I | 900 | 15 | 1:19:45 | 1:34:45 | 1 |
| 14 | Apply disinfectant to cart 1 | I | 600 | 10 | 1:34:45 | 1:44:45 | 1 |
| 15 | Initial rinse cage 1 | I | 150 | 2 | 1:44:45 | 1:47:15 | 1 |
| 16 | Partial soaping cage 1 | I | 450 | 7 | 1:47:15 | 1:54:45 | 1 |
| 17 | Final rinse cart 1 | I | 450 | 7 | 1:54:45 | 2:02:15 | 1 |
| 18 | Take cart 1 to dryer | I | 30 | 0 | 2:02:15 | 2:02:45 | 1 |
| 19 | Bring cleaning materials upstairs | I | 300 | 5 | 2:02:45 | 2:07:45 | 1 |
| 20 | Clean ceilings and walls | E | 1800 | 30 | 2:07:45 | 2:37:45 | 1 |
| 21 | Clean positioner | I | 2700 | 45 | 2:37:45 | 3:22:45 | 1 |
| 22 | Clean Cremer 2 | I | 1350 | 22 | 3:22:45 | 3:45:15 | 1 |
| 23 | Clean conveyor belts | I | 225 | 3 | 3:45:15 | 3:49:00 | 1 |
| 24 | Clean floor | E | 900 | 15 | 3:49:00 | 4:04:00 | 1 |
| 25 | Remove cleaning materials | E | 450 | 7 | 4:04:00 | 4:11:30 | 1 |
| 26 | Bring parts upstairs | I | 300 | 5 | 4:11:30 | 4:16:30 | 1 |
| 27 | Bag and label clean parts | I | 450 | 7 | 4:16:30 | 4:24:00 | 1 |
| 1 | Clear production line | I | 750 | 12 | 0:00:00 | 0:12:30 | 2 |
| 2 | Count materials | E | 600 | 10 | 0:12:30 | 0:22:30 | 2 |
| 3 | Remove materials | E | 150 | 2 | 0:22:30 | 0:25:00 | 2 |
| 4 | Bring carts | E | 150 | 2 | 0:25:00 | 0:27:30 | 2 |
| 5 | Vacuum and blow | E | 690 | 11 | 0:27:30 | 0:39:00 | 2 |
| 6 | Dismantle Cremer suction equipment | I | 450 | 7 | 0:39:00 | 0:46:30 | 2 |
| 7 | Dismantle Cremer 1 | I | 1200 | 20 | 0:46:30 | 1:06:30 | 2 |
| 8 | Dismantle T-Delta | I | 600 | 10 | 1:06:30 | 1:16:30 | 2 |
| 9 | Take carts to washing area | I | 900 | 15 | 1:16:30 | 1:31:30 | 2 |
| 10 | Insert cart 1 into washer | I | 45 | 0 | 1:31:30 | 1:32:15 | 2 |
| 11 | Initial rinse cart 1 | I | 150 | 2 | 1:32:15 | 1:34:45 | 2 |
| 12 | Soap cart 1 parts | I | 900 | 15 | 1:34:45 | 1:49:45 | 2 |
| 13 | Apply disinfectant to cart 1 | I | 600 | 10 | 1:49:45 | 1:59:45 | 2 |
| 14 | Initial rinse cage 1 | I | 150 | 2 | 1:59:45 | 2:02:15 | 2 |
| 15 | Partial soaping cage 1 | I | 450 | 7 | 2:02:15 | 2:09:45 | 2 |
| 16 | Final rinse cart 1 | I | 450 | 7 | 2:09:45 | 2:17:15 | 2 |
| 17 | Take cart 1 to dryer | I | 30 | 0 | 2:17:15 | 2:17:45 | 2 |
| 18 | Bring cleaning materials upstairs | E | 300 | 5 | 2:17:45 | 2:22:45 | 2 |
| 19 | Clean ceilings and walls | E | 1800 | 30 | 2:22:45 | 2:52:45 | 2 |
| 20 | Clean Cremer 1 | I | 1800 | 30 | 2:52:45 | 3:22:45 | 2 |
| 21 | Clean T-Delta | I | 1350 | 22 | 3:22:45 | 3:45:15 | 2 |
| 22 | Clean conveyor belts | I | 450 | 7 | 3:45:15 | 3:52:45 | 2 |
| 23 | Clean floor | E | 900 | 15 | 3:52:45 | 4:07:45 | 2 |
| 24 | Remove cleaning materials | E | 450 | 7 | 4:07:45 | 4:15:15 | 2 |
| 25 | Bring parts upstairs | I | 300 | 5 | 4:15:15 | 4:20:15 | 2 |
| 26 | Bag and label clean parts | I | 450 | 7 | 4:20:15 | 4:27:45 | 2 |
| 27 | Bag and label clean parts | I | 900 | 15 | 4:27:45 | 4:42:45 | 2 |
| 28 | Close logs and cards | E | 300 | 5 | 4:42:45 | 4:47:45 | 2 |
| 1 | Close guide | E | 200 | 3 | 0:00:00 | 0:03:20 | 3 |
| 2 | Record in production log | E | 200 | 3 | 0:03:20 | 0:06:40 | 3 |
| 3 | Change cards | E | 200 | 3 | 0:06:40 | 0:10:00 | 3 |
| 4 | Clear production line | I | 300 | 5 | 0:10:00 | 0:15:00 | 3 |
| 5 | Remove leftover material | E | 1500 | 25 | 0:15:00 | 0:40:00 | 3 |
| 6 | Access primary area | E | 300 | 5 | 0:40:00 | 0:45:00 | 3 |
| 7 | Clean ceilings and walls (secondary) | E | 2700 | 45 | 0:45:00 | 1:30:00 | 3 |
| 8 | Clean secondary equipment | I | 3600 | 60 | 1:30:00 | 2:30:00 | 3 |
| 9 | Clean technical area | E | 7200 | 120 | 2:30:00 | 4:30:00 | 3 |
| 10 | Clean secondary floor | E | 1800 | 30 | 4:30:00 | 5:00:00 | 3 |
| 11 | Close logs and cards | E | 300 | 5 | 5:00:00 | 5:05:00 | 3 |

Figure 7: Gantt Diagram of Cleaning Procedure

| Tasks | Time (s) | Frequency (1/s) | MUDA/VA | Operator |
|---|---|---|---|---|
| Change dispenser box | 50 | 0.001 | VA | OP 1 |
| Change case box | 50 | 0.0004 | VA | OP 2 |
| Change leaflet box | 50 | 0.0004 | VA | LINE COORDINATOR |
| Change label | 70 | 0.0005 | VA | LINE COORDINATOR |
| Load cases | 20 | 0.004 | VA | LINE COORDINATOR |
| Load leaflets | 40 | 0.004 | VA | LINE COORDINATOR |
| Check machine parameters | 10 | 0.0002 | MUDA | LINE COORDINATOR |
| Case quality control | 20 | 0.0022 | VA | LINE COORDINATOR |
| Record quality control result | 45 | 0.0022 | MUDA | LINE COORDINATOR |
| Monitor rejection 2 | 10 | 0.02 | VA | OP 2 |
| Discard boxes | 84 | 0.0012 | VA | OP 2 |
| Perform label rejection control (Rejection 1) | 20 | 0.01 | VA | LINE COORDINATOR |
| Pallet shipment | 180 | 0.00078 | VA | OP 2 |
| Push bottles onto the cartoner conveyor | 30 | 0.001 | MUDA | OP 2 |
| Place dispenser in case | 0.75 | 1 | VA | OP 1 |
| Label box | 1.5 | 0.2 | VA | OP 2 |
| Place cases into boxes (5 at a time) | 1 | 0.2 | VA | OP 2 |
| Palletize | 1 | 0.2 | VA | OP 2 |
| Make box | 2 | 0.2 | VA | OP 2 |

Table 4: Line Tasks Before Improvement

**CHECK-LIST DE LIMPIEZA**

Linea: BOTES PRIMARIO

OPERARIO 1

Turno:

Fecha:

| Nº | TAREA | NOMBRE OPERARIO/A | HORA INICIO | HORA FIN | TIEMPO REAL |
|---|---|---|---|---|---|
| 1 | CERRAR GUIA | | | | 3 |
| 2 | ANOTAR EN DIARIO DE PRODUCCIÓN | | | | 3 |
| 3 | TARJETAS CAMBIO COLOR | | | | 3 |
| 4 | SACAR MATERIAL SOBRANTE(ACO) | | | | 3 |
| 5 | TRAER ASPIRADORES,Y ASPIRAR( 2 PERS) | | | | 23 |
| 6 | DESMONTAR EQUIPO SUCCION CREMER ( 2PERS) | | | | 15 |
| 7 | DESMONTAR CREMER 2 (OPCIONAL) | | | | |
| 8 | DESMONTAR POSICIONADORA | | | | 10 |
| 9 | SOPLAR | | | | 10 |
| 10 | BAJAR CARROS, JAULAS AL LAVADERO | | | | 15 |
| 11 | LIMPIEZA PIEZAS LAVADERO (2PERS) | | | | 5 |
| 12 | PASAR AGUA PIEZAS | | | | 5 |
| 13 | PASAR JABON PIEZAS | | | | 30 |
| 14 | ACLARAR PIEZAS | | | | 15 |
| 15 | ECHAR DESINFECTANTE (20MIN) | | | | 20 |
| 16 | ACLARAR PIEZAS (3MIN) | | | | 5 |
| 17 | DEJAR PIEZAS EN EL SECADERO | | | | 5 |
| 18 | SUBIR MATERIAL LIMPIEZA | | | | 15 |
| 19 | LIMPIEZA TECHO Y PAREDES( 2PERS) | | | | 30 |
| 20 | LIMPIEZA POSICIONADORA | | | | 45 |
| 21 | LIMPIEZA CREMER 2 (OPCIONAL) | | | | |
| 22 | LIMPIEZA PIEZAS TRANSPORTADORA | | | | 15 |
| 23 | LIMPIEZA ESCALERAS MESA MATERIAL ADICIONAL | | | | 7 |
| 24 | LIMPIAR SUELO( 2 PERS) | | | | 30 |
| 25 | BAJAR MATERIAL DE LIMPIEZA AL LAVADERO | | | | 5 |
| 26 | SUBIR PIEZAS SECAS(EMBOLSAR Y ETIQUETAR PIEZAS LIMPIAS)( 2 PERS) | | | | 15 |

Figure 8: Standard Procedure Checklist Example